

Lecture 10

Machine Learning 1

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Overview

1 Classification

1.1 Naive Bayes Classifier

1.2 k -Nearest Neighbour

2 Introduction to Machine Learning

2.1 Types of ML Tasks

We will start this lecture by discussing two classification methods, Naive Bayes and k -Nearest Neighbour, to build on where we left off last week. Later in this lecture we will talk about more examples of classification along with different machine learning tasks.

Part 1: Classification

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Classification

Classification is a machine learning problem that requires assigning a class label to a data point.

e.g. Can we predict whether a student will pass a course or not?

training dataset		WorkGroup	Lecture	Pass
	John	F	F	F
	Peter	F	F	F
	Marie	F	T	F
	Stefan	F	T	T
	Elisa	T	F	F
	Mohammed	T	F	T
	Rachel	T	T	F
	...	T	T	T
		feature	feature	class label

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Classification is a machine learning task that involves finding the most likely class for a specific data point. Take a look at the table; each row is an instance or student, and we would like to map each student to 'T' or 'F' for the label we want to predict, which is Pass. We want the machine learning model to learn whether the student will Pass or not Pass based on the features 'workgroup' and 'lecture'. Then, given a new student for which we only know whether they attended the workgroups and lectures, the model can predict whether they will pass.

Variants of Classification Problems

Features

Usually numerical or categorical

1.1, 2.6, 1.3 vs.



Binary classification

2 classes, *for example* Pass and Enjoy Course

Multiclass classification

>2 classes, *for example* Pass and Enjoy Course and Grade

Multilabel classification

0, some, or all classes per example, *for example* class color with 7 options

1. We can have different types of features like numerical or categorical, such as having or having not attended work groups and lectures
2. Binary classification involves two classes, such as Pass *and* Enjoy Course
3. Multiclass classification has more than two classes that the classifier is trying to predict
4. Multilabel classification means that there is more than one possible label for a single class. A class called color may have red, green, blue, yellow, etc. as labels.

Classification

Prediction: what is the probability that Karla passed IS if she did not attend workgroups nor lectures?

WorkGroup	Lecture	Pass	Frequency	Probability
F	F	F	200	0.2
F	F	T	50	0.05
F	T	F	50	0.05
F	T	T	50	0.05
T	F	F	50	0.05
T	F	T	150	0.15
T	T	F	50	0.05
T	T	T	400	0.4

$$P(\text{pass} \mid \neg wg, \neg l) = \frac{P(\text{pass}, \neg wg, \neg l)}{P(\neg wg, \neg l)} = \frac{0.05}{0.25} = 0.2$$

$$P(\neg \text{pass} \mid \neg wg, \neg l) = \frac{P(\neg \text{pass}, \neg wg, \neg l)}{P(\neg wg, \neg l)} = \frac{0.2}{0.25} = 0.8$$

Note: the denominator does not depend on pass or $\neg$ pass.

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Given a new student, Karla, who has not attended the working group and has not attended the lecture, can we use probabilities to predict whether she passes the course? Recall Bayes theorem from the previous lecture:

$$P(A|B) = (P(B|A) * P(A)) / P(B)$$

We can use this theorem and plug in the numbers using the frequency of students with the two features in each row and probability of the class label in the table. We end up with a 20% probability that Karla will pass and 80% probability that Karla will not pass.

Probabilistic Classifier

We can use a probabilistic classifier to solve these types of problems. Main idea:

Input:

- Set of features (wg, l, height, age)
- Set of classes (p, not pass, male, female)
- Examples (feature values, class) (wg = True, l = False, age=20, height=183, male)

Output:

- Class C with the highest probability given the features for a *new* example

What we have just done is the basis for a probabilistic classifier. We need three things to make this work: a set of features, a set of classes, and examples with class labels for certain features. The output will be a class prediction for some new example with its own features.

Joint Probability Table *Revisited*

How to calculate

$P(\text{pass} | \text{wg}, \text{l}, \neg \text{p}, \neg \text{b}, \text{m})?$

WG	L	Pass	Understand	Bored	Gender	Pass
F	F	F	F	F	F	T
F	F	F	F	T	F	T
F	F	F	T	F	T	F
F	F	F	T	T	T	F
F	F	T	F	F	F	T
F	F	T	F	T	F	F
F	F	T	T	F	F	T
F	F	T	T	T	T	T
F	T	F	F	F	F	F
...						

How can we calculate the probability of a certain class as a conditional probability with joint features? One method is to build a Naive Bayes Classifier, which we'll look at in this next section.

Part 1.1: Naive Bayes Classifier

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Let's look at a simple probabilistic classifier: the Naive Bayes Classifier.

Towards a Bayesian Classifier 1

We can express the probability of a certain class C given that we observe features F

$$p(C | F_1, \dots, F_n)$$

Using Bayes Theorem

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

$$p(C | F_1, \dots, F_n) = \frac{p(C) p(F_1, \dots, F_n | C)}{p(F_1, \dots, F_n)}$$

In the probability lecture we saw Bayes Theorem as it appears on the left. Now, the variable A represents all the features and B represents the class. We denote the set of features like $F_1, \dots, F_n$ and the class with C .

Towards a Bayesian Classifier 2

$$p(C|F_1, \dots, F_n) = \frac{p(C)p(F_1, \dots, F_n|C)}{p(F_1, \dots, F_n)}$$

Rewrite the numerator using the chaining rule and we get

$$\frac{p(C)p(F_1|C) * p(F_2|C, F_1) * p(F_3|C, F_1, F_2) * \dots * p(F_n|C, F_1, \dots, F_{n-1})}{p(F_1, \dots, F_n)}$$

Observe that the numerator of this formula is the same for all C.

Unfortunately, this is very expensive to calculate.

In order to calculate Bayes theorem and include every single feature, we would need to use the chaining rule to rewrite the numerator into something we can work with. However, this is very complicated.

Towards a Bayesian Classifier 3

To decrease cost, Naive Bayes **assumes conditional independence of all features**

i.e. F_i is conditionally independent from F_j

$$p(F_i | C, F_j) = p(F_i | C)$$

Resulting in new formula:

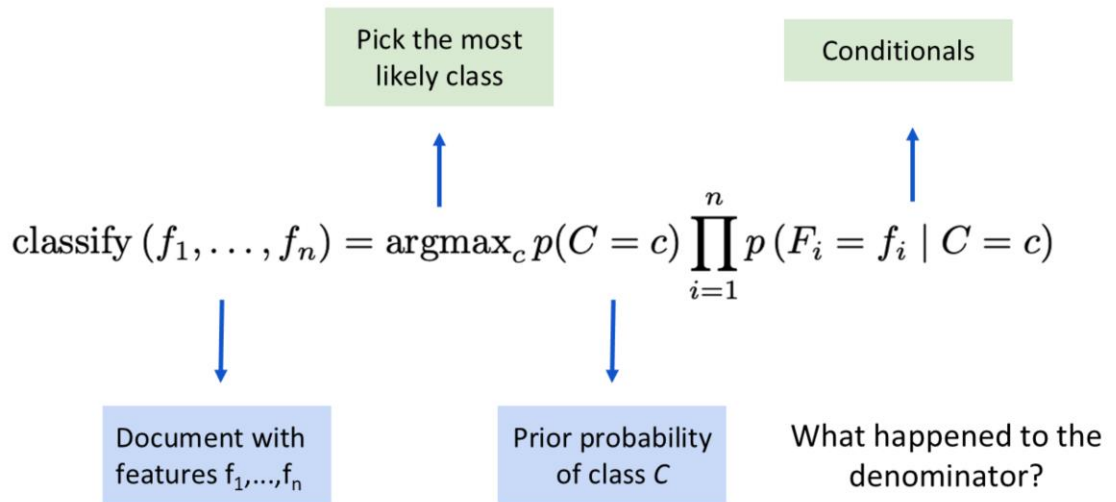
$$\begin{aligned} p(C | F_1, \dots, F_n) &= p(C) * p(F_1 | C) * p(F_2 | C) \dots \\ &= p(C) \prod p(F_i | C) \end{aligned}$$

Assumption of conditional independence is why the classifier is called **naive**.

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What can we do instead? The *Naive* Bayes Classifier assumes conditional independence of all the features. This means, for example, that the likelihood of a student attending the lecture does not affect the probability of a student attending a workgroup. This assumption is useful because we can rewrite the formula into something much simpler.

Naive Bayes Classifier



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Now we have this new formula which we can use to make a classifier.

Naive Bayes Classifier

The Naive Bayes classifier may perform well even in cases when conditional independence assumptions are incorrect.

Classification problems require predicting the right class label, *not* accurately estimating probabilities.

Correct estimation $\Rightarrow$ accurate prediction

Accurate prediction $\nRightarrow$ correct estimation

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The conditional independence assumption does not mean that our predictions are no longer useful. The goal of a probabilistic classifier is to make a good prediction of which class an instance belongs to, not estimating the exact values.

Naive Bayes *Example*

Let's build a classifier that decides whether a sentence belongs to the course Intelligent Systems or Introduction to Programming.

Step 1: Choose features, *e.g.* words

- Python
- List
- Algorithm
- Boolean
- File
- Search
- Programming
- Probability

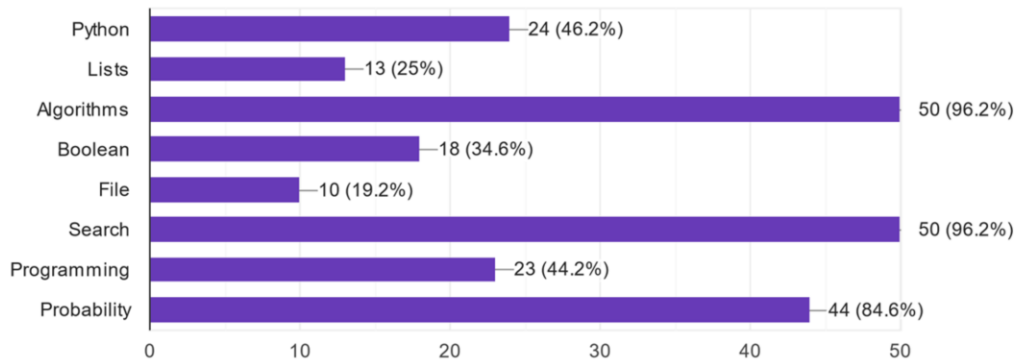
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Let's build a Naive Bayes Classifier from scratch. Let's say we want our classifier to predict whether a given sentence belongs to the course IS or Programming. The first step is to choose the features of the sentence we care about; in this case we choose the occurrence of words. Note that we could have also chosen features such as sentence length, number of total words, etc., but this is less useful considering our goal.

Intelligent Systems Keywords

Choose up to 5 terms that describe the Intelligent Systems course.

52 responses



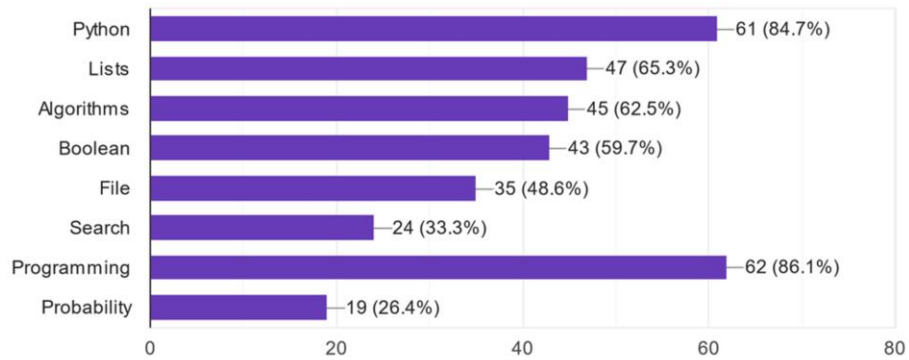
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In order to make these features useful for our probabilistic classifier, we need to know how *probable* each word is for a given class. Shown here are the results for IS from the votes from a class in a previous year...

Programming Keywords

Programming

72 responses



... and these are the results for Programming.

Naive Bayes *Example*

Step 2: Training

- Count occurrences of words
- Split into train/test datasets

Feature	Prog.	Int. Sys
Python	61	24
Programming	62	23
Probability	19	44
Algorithm	45	50
Lists	47	10
Boolean	43	18
File	35	10
Search	24	50
TOTAL	72	52

We can use this data to make a table with total votes for each word for each class.

Naive Bayes *Example*

Step 2: Training

- Class probabilities
- Conditional probabilities
- Smoothing (simple)

Probability	C = IPP	C = IS
P(IS)		42%
P(IPP)	58%	
P(Python C)	84%	46%
P(Programming C)	86%	44%
P(Probability C)	26%	84%
P(Algorithm C)	62%	96%
P(List C)	65%	25%
P(Boolean C)	59%	34%
P(File C)	48%	19%
P(Search C)	33%	96%

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Which then becomes probabilities of each feature for each class. Note that the first two rows represent the base probability of a sentence belonging to each class, regardless of its features.

Naive Bayes Example

Step 3: Calculate probability for each class given a new example with features **python, file, lists**.

$P(\text{IS} | \text{Python, File, Lists})$

$$\begin{aligned} &= P(\text{IS}) * P(\text{Python} | \text{IS}) * P(\text{File} | \text{IS}) * P(\text{Lists} | \text{IS}) \\ &= 0.42 * 0.46 * 0.19 * 0.25 = 0.009 \end{aligned}$$

Why is the sum not 1?

$P(\text{IPP} | \text{Python, File, Lists})$

$$\begin{aligned} &= P(\text{IPP}) * P(\text{Python} | \text{IPP}) * P(\text{File} | \text{IPP}) * P(\text{Lists} | \text{IPP}) \\ &= 0.58 * 0.84 * 0.48 * 0.65 = 0.15 \end{aligned}$$

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Let's say our new sentence with the words python, file, and lists. Using the chaining rule we now extend the Bayes formula, multiplying only the probabilities for the words which occur in the new sentence. The results show that there is a 0.009 probability that a sentence with 'python', 'file', and 'lists' belongs to IS, and 0.15 probability that it belongs to Programming. Since $0.15 > 0.009$ we would classify this sentence as belonging to the Programming class. Note that the sum of these probabilities is not 1 because we removed the denominator of Bayes Theorem.

Smoothing

How do we make sure the product of our formula doesn't become zero when the probability of one word is zero?

Smoothing:

- **Option A:** Add 1 example for each value to your table
- **Option B (Laplace):** Make the value 0.01 if $P(F|X)=0$.

If not specified, choose **option B**.

Probability	C = IPP	C = IS
P(IS)		42%
P(IPP)	58%	
P(Python C)	84%	46%
P(Programming C)	86%	44%
P(Probability C)	26%	84%
P(Algorithm C)	62%	96%
P(List C)	65%	25%
P(Boolean C)	59%	34%
P(File C)	48%	19%
P(Search C)	33%	96%
P(Stefan C)	0%	100%

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Imagine now that we have a certain feature, in this case the word Stefan, for which there is a 0% probability of it belonging to a class. This is a problem because once we plug it into our Bayes Theorem, the entire formula will multiply out to 0 regardless of the other features. To solve this issue we have two options: add one example for each value to your table of *occurrences*, or we can perform **Laplace smoothing**, where we make the value = 0.01. This value is small enough to convey that it is very unlikely that this feature occurs for this class, but prevents the formula from becoming 0.

Naive Bayes Classifier *Example Recipe*

1. Preprocessing

- Choose features, *e.g.* words
- Split data into training (*e.g.* $\frac{2}{3}$) and test (*e.g.* $\frac{1}{3}$) sets

2. Training

- Count occurrences of words
- Simple Smoothing (+1)
- Prior probabilities
- Conditional probabilities

3. Testing

- Apply Naive Bayes to find class with maximal probability

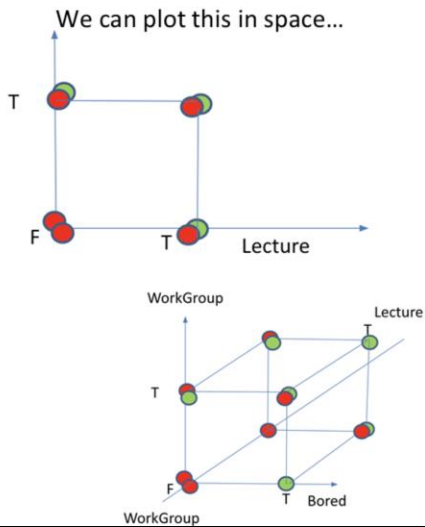
Part 1.2: k-Nearest Neighbour

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K-Nearest Neighbor is another simple. classification algorithm.

Joint Probability Table *Revisited*



	WorkGroup	Lecture	Pass
John	F	F	F
Peter	F	F	F
Marie	F	T	F
Stefan	F	T	T
Elisa	T	F	F
Mohammed	T	F	T
Rachel	T	T	F
...	T	T	T

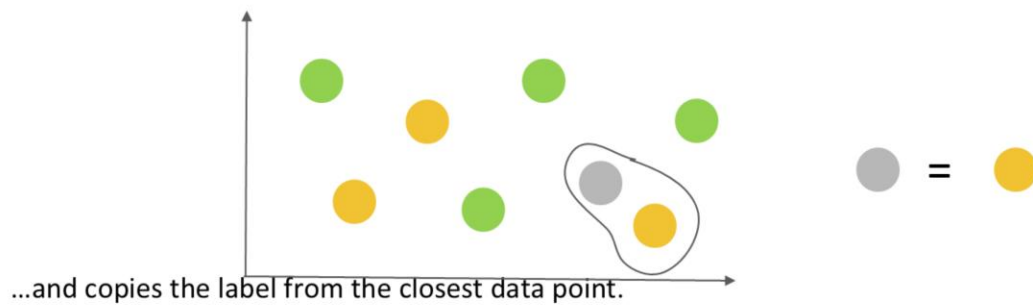
This works with any continuous variables,
e.g. age, hours spent.

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The important concept here is to realise that we can turn any features in a table into points in space. This is useful because then we can talk about concepts like the distance between two points. This can be a 2D space (square), 3D space (cube), etc.

Nearest Neighbour

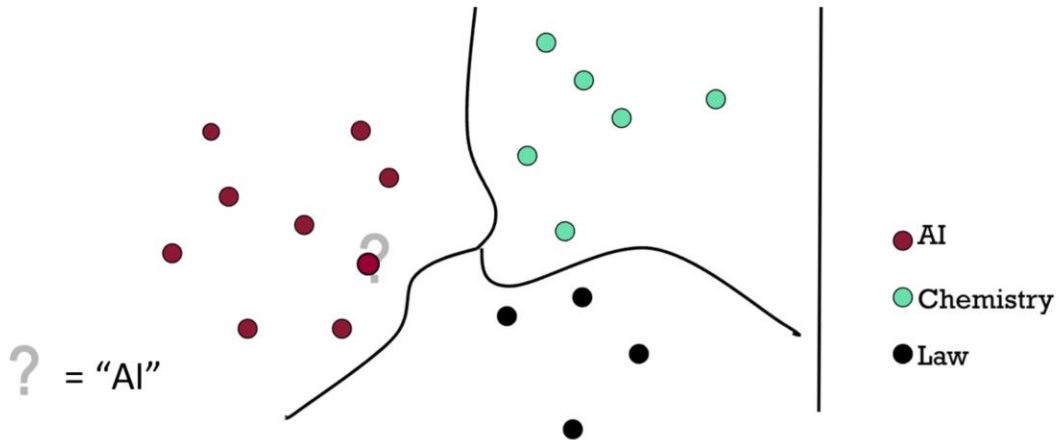
Nearest Neighbour finds the distance between a new example and all the examples in the data which have been labeled already.



Working in 2D space, we can use the Nearest Neighbor algorithm to predict the class for a new data point. The rule is simple: find the labeled data point that is the nearest in distance and copy its label. Here the closest example to the gray point is yellow, thus we copy the gray label for this point.

Nearest Neighbour Example

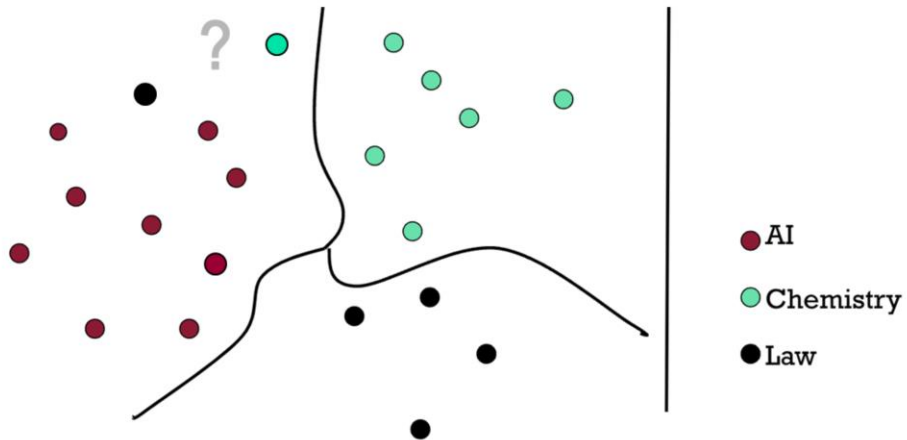
Classify a piece of text according to a set of documents belong to class AI, Chemistry, or Law.



Here it is easy to identify the new point as the AI class.

Nearest Neighbour Example

This method can't handle any noise in the data...



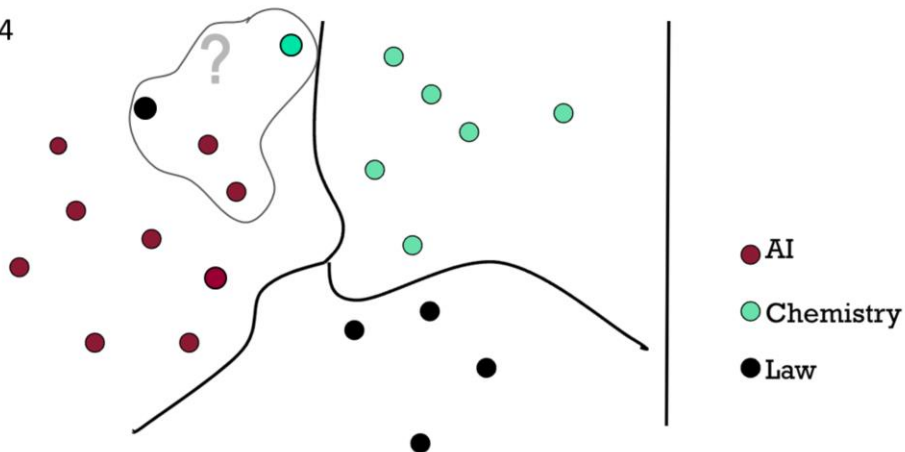
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But this method does not handle noisy data well. The problem is that we would identify the new data point as chemistry instead of AI.

Solution: k -Nearest Neighbour

Find k most similar cases and copy the majority class.

e.g. $k = 4$



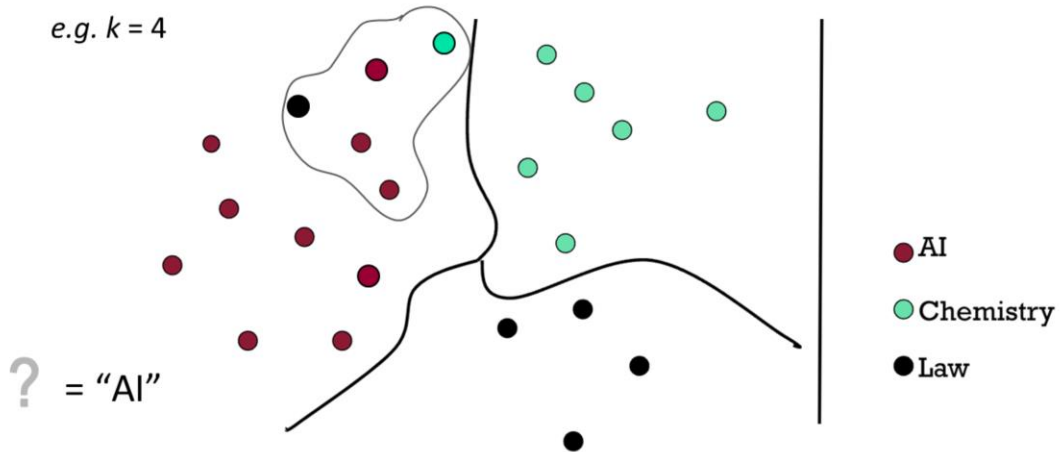
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New idea: instead of choosing the single nearest neighbour we look at the majority class label of the k -nearest neighbours. K is a value that is defined by the data engineer, in this case $k=4$, thus we count the majority class of the 4 nearest neighbours and copy the label. If we are faced with a tie, e.g. we have two AI points and two chemistry points, we may choose to extend k to 5.

Solution: k -Nearest Neighbour

Find k most similar cases and copy the majority class.

e.g. $k = 4$



Part 2: Introduction to Machine Learning

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AI creates gallery of nightmare images for Halloween

25 October 2016 Technology

Share



The nightmare machine turned pictures from the second presidential debate into darker visions - the images have an impressionistic, sketchy quality because of the way the AI translates them.

MIT

THE SHIFT

An A.I.-Generated Picture Won an Art Prize. Artists Aren't Happy.

"I won, and I didn't break any rules," the artwork's creator says.

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UK ► UK politics Education Media Society Law Scotland Wales Northern Ireland

Police

Facial recognition could transform policing in same way as DNA, says Met chief

CNN BUSINESS

Audio Live TV

Log In

Regulators give green light to driverless taxis in San Francisco

By Nadia Bidarian, CNN

Published 6:32 AM EDT, Fri August 11, 2023

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Machine learning has become very popular in the last decade. Many of the techniques we use today have been around for 30 years, but computers and graphic cards have gotten much faster in recent years, enabling us to use those techniques efficiently in practice.

Adaptive Intelligent Agents

Adaptive intelligent agents respond to their environment or other agents and **learn** from these interactions over time.

Learning

- Modifies the agent's decision mechanisms
- Is essential for unknown environments
- Useful as a system construction method



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Intelligent systems are *agents*: things in a world, that sense the world and execute actions. Now *adaptive* agent, an agent that learns from past experience to improve its future behaviour. It's an agent that can avoid past mistakes, and repeat past successes.

In the example of the floor cleaning robot, this would be a robot that could slowly learn to recognise your house, and the most efficient way to navigate it. It might learn when you're likely to be out and choose those times to vacuum. If it responds to voice commands it might get better and better at recognising your voice, and learn to tune out the background noise of your house.

Learning Movie Recommendations

Close

Other Movies You Might Enjoy

[Amélie](#)



Add

★★★★☆

Not Interested

[Y Tu Mama Tambien](#)



Add

★★★★☆

Not Interested

[Guys and Girls](#)



Add

★★★★☆

Not Interested

[Mostly Martha](#)



Add

★★★★☆

Not Interested

[Only Human](#)



Add

★★★★☆

Not Interested

[Russian Dolls](#)



Add

★★★★☆

Not Interested



Eiken has been added to your Queue at position 2.

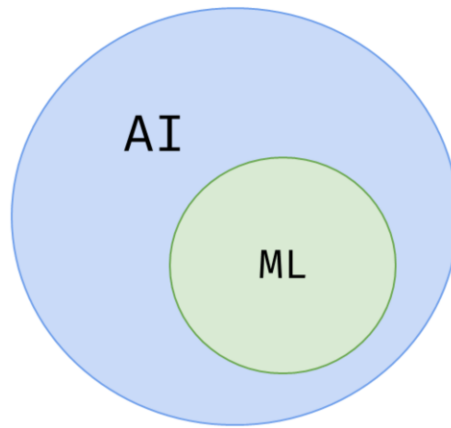
This movie is available now.

[Move To Top Of My Queue](#)

[Continue Browsing](#) [Visit your Queue](#)

The way Netflix presents you with movies and TV shows on your homepage is another example of machine learning. These recommendations are also personalised on shopping sites and other platforms that know about your preferences.

Machine Learning vs. Artificial Intelligence

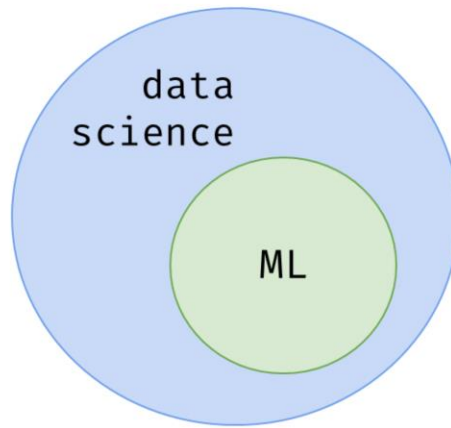


AI but not ML: chess algorithm via Search, automated reasoning, planning.

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Machine learning and AI are not synonymous. Many AI technologies are not actually based on learning; many of these methods have been around for a long time. For example, the adversarial search methods that we've looked at in this course already. Machine learning requires *learning and adaptivity* elements.

Data Science vs. Artificial Intelligence

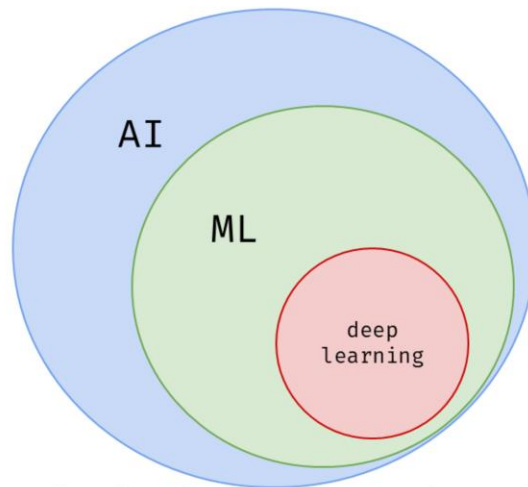


Data science but not ML: gathering data, harmonising and interpreting data

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Data science is a larger practice that can include AI and machine learning but is more about the process of gathering, cleaning, and interpreting data.

Deep Learning vs. ML vs. AI



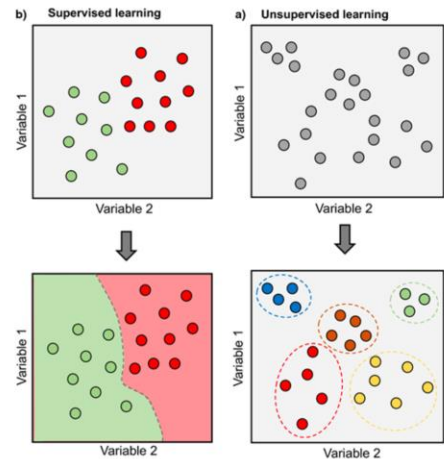
Deep learning is a type of ML based on neural networks, modeled similarly to the human brain.

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Deep learning is a subset of ML and AI that has shown a lot of promise and progress in the last several years. Deep learning is the basis of large systems you are familiar with, for example Chat GPT and DALLe.

Types of Learning

1. **Supervised**: training data includes correct label for each example
 - a. May have only positive examples or positive and negative
2. **Unsupervised**: training data does not include labels
 - a. The model must find the hidden structures in examples by itself



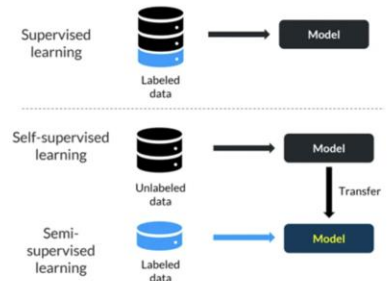
There are four major types of learning when we talk about machine learning. The first is **supervised learning**, in which we give the model *labeled* data. These *labels* give the algorithm some information about each data point. In the diagram you can see that the supervised learning method has each data point from the training set labeled as either green or red. **Unsupervised learning** uses training data which is *unlabeled*. In the example, the training data is not labeled as either green or red to begin with, and it's up to the algorithm to decide how to group these data points together.

https://www.researchgate.net/figure/Supervised-and-unsupervised-machine-learning-a-Schematic-representation-of-an_fig3_351953193

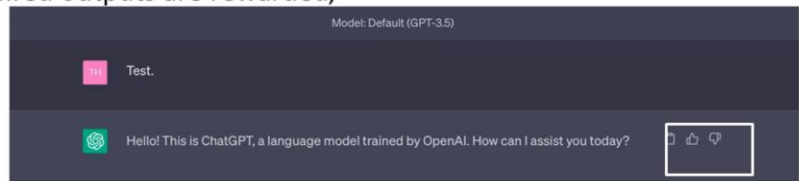
Types of Learning

3. **Semi-supervised**: partially labeled examples

- a. Active learning: label only high-quality data that improves model performance



4. **Reinforcement**: desired outputs are rewarded, undesired are punished



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Thirdly we have **semi-supervised** learning, a combination of supervised and unsupervised learning. **Reinforcement learning** is slightly different from the other three, in which the algorithm knows the 'correct' answers through reward or punishment. For example, when you use the thumbs-up or thumbs-down features of ChatGPT, you are helping to train the algorithm by rewarding or punishing the answer it has given. We will focus more on the first three types of learning in this course.

Part 2.2: Types of ML Tasks

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Types of Learning Problems / Tasks

Within learning types, there are different problems that require different algorithms and give different outputs. Main ML problems include

Supervised	Classification
	Regression
	Ranking
	Collaborative filtering
Unsupervised	Clustering

Within each type of learning there are different types of algorithms that solve problems. We will talk a lot about **classification** and **regression** in this course.

Classification

Type of learning: *supervised*.

Finds the most likely class for each instance.

e.g. handwriting recognition



MNIST

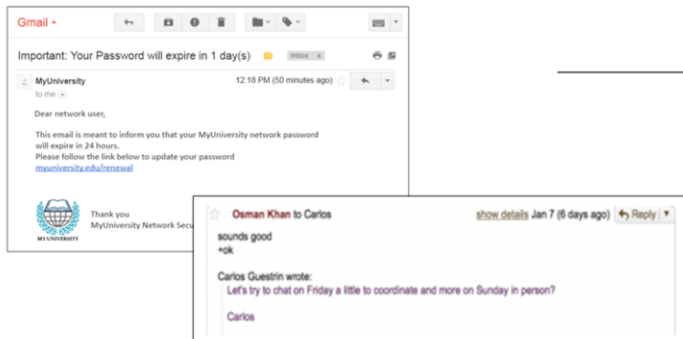


Classification is a machine learning task in which the algorithm chooses the most likely class that a certain instance belongs to. For example, given a picture of some handwriting on this letter, the algorithm must choose the class that each symbol belongs to, in which the class is a letter A-Z or a number 0-9. The MNIST dataset is a massive dataset with 6000 examples of handwriting which can be used to train a model to identify the correct class for a certain symbol. The right picture shows a collection of these examples for the number '2'.

Classification Example

e.g. spam filtering

DATA



PREDICTION

spam
vs.
not spam

Another example of classification is deciding whether a given email belong to the class 'spam' or 'not spam'. An algorithm makes this decision based on different *features* of the email, for example whether there are a lot of capital letters or misspelled words.

Classification *Example*

e.g. facial recognition



Example training images
for each orientation



Facial recognition is the classification of a facial image to a specific individual. Each photo of your face is an instance of the class that is you.

Classification Example

e.g. ALVINN (1995)

.3	...	.3	right
.0	...	.0	left
.0	...	.0	left
.1	...	.2	right
.3	...	.4	straight
.1	...	.6	left
.2	...	.3	right
.4	...	.4	left
.6	...	.6	straight
.3	...	.3	straight
.0	...	.0	left



< 30 x 32 pixels

960 features



Autonomous driving also involves classification. This is an early example of autonomous driving with the ALVINN system of 1995. Each image that is captured by the camera on the vehicle corresponds to an action made by the driver. In this way, the algorithm can map a certain image of the road to an action, such as 'turn left' or 'brake'. In today's systems, each object in the screen is classified as 'person' or 'stop sign', etc.

Classification *Example*

e.g. weather prediction



Weather prediction systems choose which weather condition (class) is most likely for a certain weather pattern (instance).

Regression

Type of learning: *supervised*. Predict a numeric value for a data point.

e.g. stock market prediction

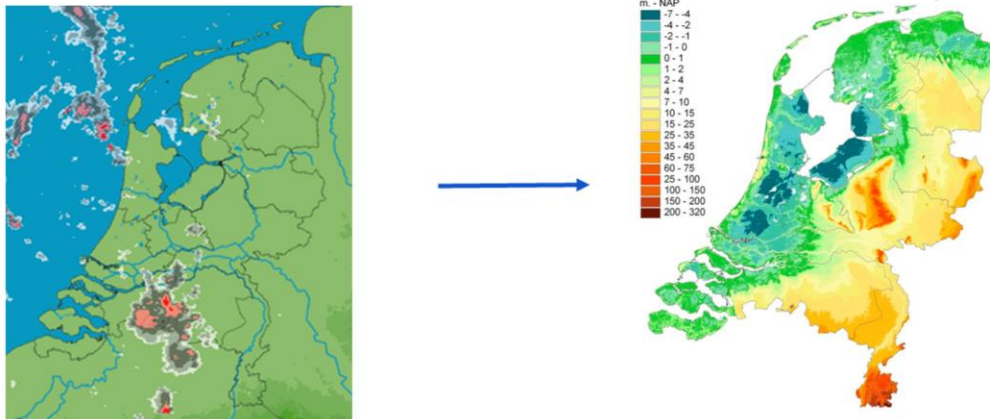


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In regression, the goal is to predict a specific *number* instead of a *class*. One example is stock market prediction, in which an algorithm tries to predict stock value, in numeric form, by fitting a line to the graph above.

Regression Example

e.g. weather prediction revisited



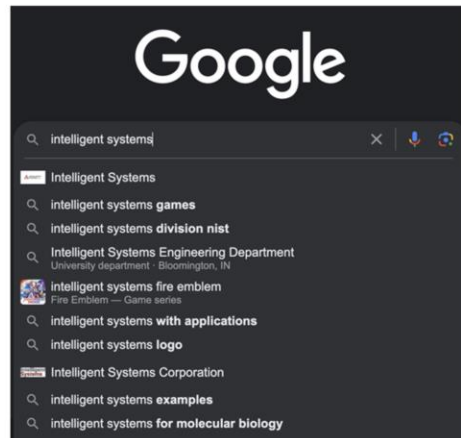
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Let's revisit our weather example. Instead of predicting a certain weather condition out of a set list of possible weather conditions, regression is used to predict a specific number, such as temperature. Other regression tasks could be to predict air pressure or wind speed.

Ranking

Type of learning: *supervised*. Compare items in order to rank them by a certain criteria.

e.g. web search results

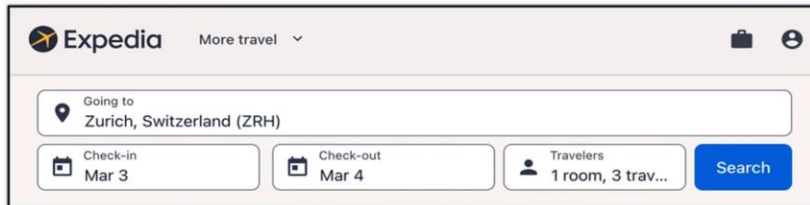


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Next we have ranking, which is most common in online platforms that rank content for our consumption. For example, Google ranks the web search results of your query when you use their site. They use data from the past in which they learned that users were happy with the results, for example by knowing which sites users stayed on the longest.

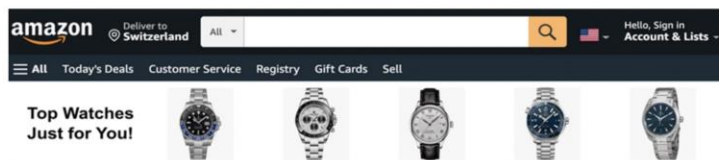
Ranking Example

e.g. travel agency results



The image shows the Expedia website's search interface. At the top is the Expedia logo and a 'More travel' dropdown menu. Below this is a search bar with a location pin icon and the text 'Going to Zurich, Switzerland (ZRH)'. To the right of the search bar are three input fields: 'Check-in Mar 3', 'Check-out Mar 4', and 'Travelers 1 room, 3 trav...'. A blue 'Search' button is located to the right of the 'Travelers' field.

e.g. digital marketplaces

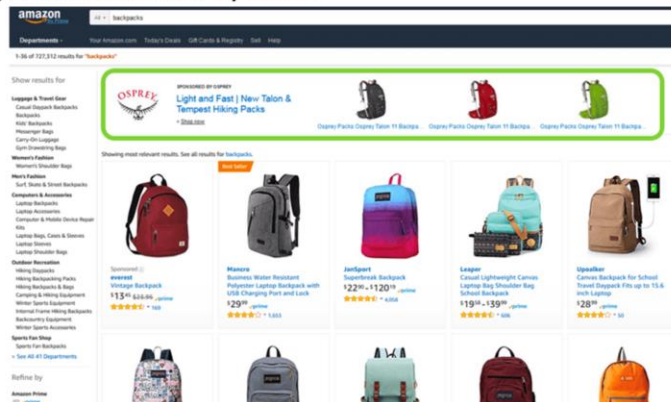


More examples of ranking include travel agency results and digital marketplaces like Amazon.

Collaborative Filtering

Type of learning: *(un)supervised*. Filters items that a user might like based on interests of similar users.

e.g. recommender systems

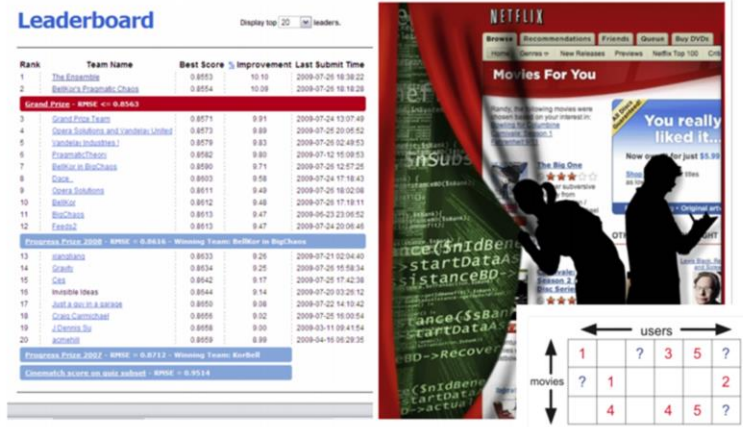


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Collaborative filtering systems are used in a lot of the same platforms as ranking, but specifically works by comparing your preferences with the profiles of similar users. It is distinct from ranking because it involves the use of your specific data attached to your profile, for example when you are logged in to a platform. The profile of each user is built up from data such as their previous searches and purchases. Collaborative filtering can either be supervised or unsupervised depending on the specific method.

Collaborative Filtering Example

Netflix ML competition 2006/09 with \$1 million prize.

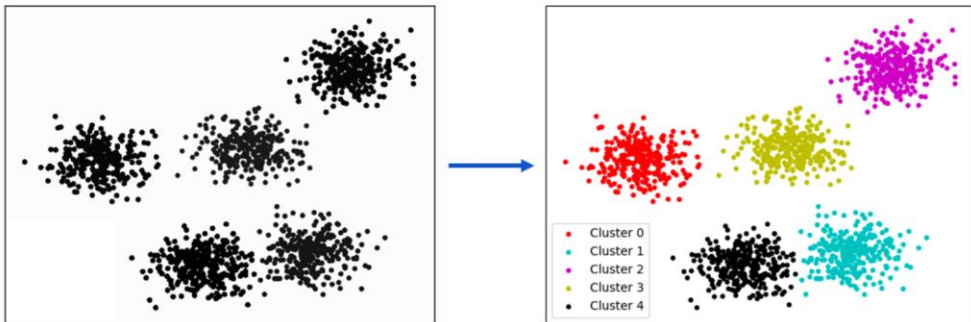


Platforms like Netflix also use collaborative filtering using your personal data.

Clustering

Type of learning: *unsupervised*. Discovering structure in the data to group similar items.

- Requires a criteria for similarity



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Clustering is a type of unsupervised learning (recall, this means the training data is *unlabeled*), used to discover similar structure between data points. For example, given the following data points, which similarities can we find to group them into clusters?

Clustering *Example*

e.g. image grouping



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Clustering can also be applied to a large collection of images in order to group them into different categories. Since the algorithm does not have any labeled training data, it finds data structures to use in order to cluster images, for example color and intensity, shapes, image quality, etc.

Clustering *Example*

e.g. grouping similar documents



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Similarly, clustering algorithms can perform the same function as a librarian, choosing which documents or books should be clustered into similar categories.

Intelligent Systems Framework

Efficient, simple, correct, well- understood, generic	Find best decision based on experience.	Weeks 6 & 7
	Find best decision with knowledge & reason.	Weeks 3, 4, 5
	Find best decision in multiple agent setting.	Weeks 1 & 2
	Find best decision with heuristics.	
	Systematically find all possible solutions.	

Progress in the intelligent systems framework.

Summary

Take Home Message

Machine learning allows us to use probabilities to approximate answers and make **predictions**. Different types of **tasks** can deal with different data modalities and outputs.

The four major types of learning specify how we use labels on **training data**, and the five major problem types differ in the predictions they make.

Within classification, we can use **Naive Bayes** to find the most likely class for a data point, or use **k-Nearest Neighbour** to find the majority class of a data point's neighbours.

Thank you for your attention and have a great day.